



Migraine

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Migraine is an episodic primary headache disorder. Symptoms typically last 4 to 72 hours and may be severe. Pain is often unilateral, throbbing, worse with exertion, and accompanied by symptoms such as nausea and sensitivity to light, sound, or odors. Auras occur in about 25% of patients, usually just before but sometimes after the headache. Diagnosis is clinical. Treatment is with triptans, [dihydroergotamine](#), antiemetics, and analgesics. Preventive regimens include lifestyle modifications (eg, sleep habits or diet) and medications (eg, beta-blockers, [amitriptyline](#), [topiramate](#), divalproex, monoclonal antibodies).

(See also [Approach to the Patient With Headache](#).)

Epidemiology of Migraine

Migraine is the most common cause of recurrent moderate to severe headache; 1-year prevalence is 18% for women and 6% for men in the United States ([1](#)). Migraine most commonly begins during puberty or young adulthood, waxing and waning in frequency and severity over the ensuing years; it often diminishes after age 50. Studies show familial aggregation of migraine.

Evidence suggests that migraine may frequently develop after mild [traumatic brain injury](#) ([2,3](#)).

Epidemiology references

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- [Ishii R, Schwedt TJ, Trivedi M, Dumkrieger G, Cortez MM, Brennan KC, Digre K, Dodick DW.](#) Mild traumatic brain injury affects the features of migraine. *J Headache Pain*. 2021 Jul 22;22(1):80. doi: 10.1186/s10194-021-01291-x
- [Couch JR, Stewart KE.](#) Headache Prevalence at 4-11 Years After Deployment-Related Traumatic Brain Injury in Veterans of Iraq and Afghanistan Wars and Comparison to Controls: A Matched Case-Controlled Study. *Headache*. 2016;56(6):1004-1021. doi:10.1111/head.12837

Pathophysiology of Migraine

Migraine is thought to be a neurovascular pain syndrome with altered central neuronal processing (activation of brain stem nuclei, cortical hyperexcitability, and spreading cortical depression) and involvement of the trigeminovascular system (triggering neuropeptide release [eg, CGRP], which causes painful inflammation in cranial vessels and the dura mater).

Many potential migraine triggers have been identified; they include the following (1):

- Drinking red wine
- Skipping meals
- Excessive afferent stimuli (eg, flashing lights, strong odors)
- Weather changes
- Sleep deprivation
- Stress
- Hormonal factors, particularly menstruation
- Certain foods

Food triggers vary from person to person.

[Head trauma](#), [neck pain](#), or [temporomandibular joint dysfunction](#) sometimes triggers or exacerbates migraine.

Fluctuating estrogen levels are a potent migraine trigger. Many women have onset of migraine at menarche, severe attacks during menstruation (menstrual migraine), and worsening during menopause. For most women, migraines remit during pregnancy (but sometimes worsen during the first or second trimester); they worsen after childbirth, when estrogen levels decrease rapidly.

Oral contraceptives and other hormone therapy occasionally trigger or worsen migraine and have been associated with stroke in women who have migraine with aura.

Familial hemiplegic migraine, a rare subtype of migraine, is associated with genetic defects on chromosomes 1, 2, and 19. The role of genes in the more common forms of migraine is under study (2). In some families, the migraine phenotype varies considerably, causing primarily headache in some family members, vertigo in others, and hemiplegia or an aura in others. This finding suggests that migraine may actually be a more generalized disorder and not just a headache disorder.

Pathophysiology references

1. [Kelman L](#). The triggers or precipitants of the acute migraine attack. *Cephalalgia*. 2007 May;27(5):394-402. doi: 10.1111/j.1468-2982.2007.01303.x.
2. [Grangeon L](#), [Lange KS](#), [Waliszewska-Prosół M](#), [Onan D](#), [Marschollek K](#), [Wiels W](#), [Mikulénka P](#), [Farham F](#), [Gollion C](#), [Ducros A](#); [European Headache Federation School of Advanced Studies \(EHF-](#)

[SAS](#)). Genetics of migraine: where are we now? *J Headache Pain*. 2023 Feb 20;24(1):12. doi: 10.1186/s10194-023-01547-8

Symptoms and Signs of Migraine

Often, a **prodrome** (a sensation that a migraine is beginning) heralds attacks. The prodrome may include mood changes, neck pain, food cravings, loss of appetite, nausea, or a combination.

An **aura** precedes attacks in about 25% of patients. Auras are temporary neurologic disturbances that can affect sensation, balance, muscle coordination, speech, or vision; they last minutes to an hour. The aura may persist after headache onset. Most commonly, auras involve visual symptoms (fortification spectra—eg, binocular flashes, arcs of scintillating lights, bright zigzags, scotomata). Paresthesias and numbness (typically starting in one hand and marching to the ipsilateral arm and face), speech disturbances, and transient brain stem dysfunction (eg, causing ataxia, confusion, or even obtundation) are less common than visual auras. Some patients have an aura with little or no headache.

Headache varies from moderate to severe, and attacks last from 4 hours to several days, typically resolving with sleep. The pain is often unilateral but may be bilateral, most often in a frontotemporal distribution, and is typically described as pulsating or throbbing.

Migraine is more than a headache. Associated symptoms such as nausea (and occasionally vomiting), photophobia, sonophobia, and osmophobia are prominent. Patients report difficulty concentrating during attacks. Routine physical activity usually aggravates migraine headache; this effect, plus the photophobia and sonophobia, encourages most patients to lie in a dark, quiet room during attacks. Severe attacks can be incapacitating, disrupting family and work life.

Attacks vary significantly in frequency and severity. Many patients have several types of headache, including milder attacks without nausea or photophobia that may resemble [tension-type headache](#) but are a forme fruste of migraine.

Chronic migraine

Patients with episodic migraine can develop chronic migraine (previously termed combination or mixed headache) with features of migraine and tension-type headache. These patients have headaches ≥ 15 days/month. Overuse headache often contributes to the conversion of episodic to chronic migraine.

Other symptoms

Rare forms of migraine can cause other symptoms:

- **Migraine with brain stem aura** (previously called basilar artery migraine) causes combinations of vertigo, ataxia, visual field loss, sensory disturbances, focal weakness, and altered level of consciousness.
- **Hemiplegic migraine**, which may be sporadic or familial, causes unilateral weakness.

Diagnosis of Migraine

- History and physical examination

Diagnosis of migraine is based on characteristic symptoms and a normal physical examination, which includes a thorough [neurologic examination](#).

Red flag findings that suggest an alternate diagnosis (even in patients known to have migraine) include the following:

- Pain that reaches peak intensity within a few seconds or less (thunderclap headache)
- Onset after age 50
- Headaches that increase in intensity or frequency for weeks or longer
- History of cancer (brain metastases) or an immunosuppressive disorder (eg, HIV infection)
- Fever, meningismus, altered mental status, or a combination
- Persistent focal neurologic deficits
- [Papilledema](#)
- A clear change in an established headache pattern

Patients with characteristic symptoms and no red flag findings do not require testing. Patients with red flag findings often require [testing](#), including MRI and sometimes lumbar puncture.

Common diagnostic errors include the following:

- Not realizing that migraine often causes bilateral pain and is not always described as throbbing
- Misdiagnosing migraine as sinus headache or eye strain because autonomic and visual symptoms of migraine are absent
- Assuming that any headache in patients known to have migraine represents another migraine attack (a thunderclap headache or a change in the previous headache pattern may indicate a new, potentially serious disorder)
- Mistaking migraine with aura for a transient ischemic attack, especially when the aura occurs without headache, in older adults
- Diagnosing a thunderclap headache as migraine because a triptan relieves it (a triptan can also relieve a headache due to subarachnoid hemorrhage)

Several unusual disorders can mimic migraine with aura:

- Dissection of the carotid or vertebral artery
- Cerebral vasculitis
- Moyamoya disease
- CADASIL (cerebral autosomal dominant arteriopathy with subcortical infarcts and leukoencephalopathy)

- [MELAS](#) (mitochondrial encephalopathy, lactic acidosis, and strokelike episodes) syndrome

Treatment of Migraine

Migraine treatment may include the following approaches ([1](#), [2](#)):

- Elimination of obvious triggers
- Relaxation techniques, yoga, or behavioral interventions
- For mild headaches, [acetaminophen](#) or nonsteroidal anti-inflammatory drugs (NSAIDs)
- For acute attacks, triptans, [lasmiditan](#), gepants (small-molecule [calcitonin](#) gene-related peptide [CGRP] receptor antagonists), or [dihydroergotamine](#) plus a dopamine antagonist antiemetic
- Neuromodulatory devices for acute treatment and prevention

A thorough explanation of the disorder helps patients understand that although migraine cannot be cured, it can be controlled, enabling them to better participate in treatment.

Patients are urged to keep a written headache diary to document the number and timing of attacks, possible triggers, and response to treatment. Identified triggers are eliminated when possible. However, triggers are often overdiagnosed and should be avoided in moderation; elimination of all possible triggers can be excessive and lead to unnecessary life restrictions.

Choice of medications used to treat acute migraine headache is based on frequency, duration, and severity of attacks. Analgesics, antiemetics, triptans, [lasmiditan](#), gepants, or [dihydroergotamine](#) may be used ([3](#)). If patients wish to avoid medications or if medications have been ineffective, [neuromodulatory treatments](#) can sometimes be used for acute attacks and/or prevention.

Patients who frequently (eg, > 2 days/week) use medications to treat their acute migraine attacks (particularly analgesics that contain butalbital, triptans, [ergotamine](#), or opioids) should be treated with preventive migraine drugs combined with a program for stopping overused analgesics.

Clinicians sometimes recommend behavioral interventions (biofeedback, stress management, psychotherapy) to manage migraine, especially when stress is a major trigger or when analgesics are being overused.

Yoga can reduce headache frequency and intensity; it enhances vagal tone and decreases sympathetic drive, thus improving cardiac autonomic balance. Relaxation techniques can reduce sympathetic nervous system activity, ease muscle tension, and alter brain wave activity.

Acute attacks

Initial treatment of newly diagnosed mild to moderate migraine attacks is with NSAIDs or [acetaminophen](#). For patients who do not respond adequately, clinicians should add a triptan ([3](#)).

If these drugs are ineffective, [dihydroergotamine](#), or gepants (small-molecule [calcitonin](#) gene-related peptide [CGRP] receptor antagonists) may be used. A good response to a triptan, [dihydroergotamine](#), or a gepant should not be interpreted as diagnostic for migraine because these medications may relieve headache due to subarachnoid hemorrhage or other structural abnormalities.

If mild attacks worsen or if attacks are severe from the onset, triptans or [dihydroergotamine](#) can be used. When nausea is prominent, combining a triptan with an antiemetic at the onset of attacks is effective.

Triptans are selective serotonin 1B,1D receptor agonists. They are not analgesic per se but specifically block the release of neuropeptides that trigger migraine pain. Triptans are most effective when taken at the onset of attacks. They are available in oral, intranasal, and subcutaneous forms; subcutaneous forms are more effective but have more adverse effects. Overuse of triptans can lead to [medication overuse headache](#). Triptans and [dihydroergotamine](#) can cause coronary artery constriction and are thus contraindicated in patients with coronary artery disease or uncontrolled hypertension; these medications must be used with caution in older patients and in patients with vascular risk factors.

[Lasmiditan](#) (a new selective serotonin [5-HT] 1F receptor agonist) or a gepant, such as [ubrogepant](#) or [rimegepant](#), can be used when triptans or [dihydroergotamine](#) are contraindicated because of cardiovascular disorders. [Lasmiditan](#), which has a much greater affinity for serotonin 1F receptors than for 1B receptors, has no cardiovascular contraindications. (Triptans cause vasoconstriction by activating 5-HT1B receptors.)

Gepants are oral CGRP antagonists. Unlike other acute migraine medications, they do not cause medication overuse headaches and can be used for migraine prevention. Gepants have no known serious cardiovascular or gastrointestinal effects and no cardiovascular contraindications.

An antiemetic (eg, [metoclopramide](#), [prochlorperazine](#)) alone may be used to relieve mild or moderate attacks. [Prochlorperazine](#) suppositories (25 mg) or tablets (10 mg) are an option for patients who cannot tolerate triptans and other vasoconstrictors.

Evidence supports use of neuromodulatory devices for acute attacks and prevention of migraine headaches ([4,5](#)).

Intractable attacks

Hydration with intravenous fluids, such as normal saline, can help relieve headache and increase a sense of well-being, especially in patients who are dehydrated from vomiting.

Severe persistent attacks can be treated with a combination of intravenous [dihydroergotamine](#) and a [dopamine](#) antagonist antiemetic (eg, [metoclopramide](#) or [prochlorperazine](#)). [Dihydroergotamine](#) is also available in a subcutaneous form and as a nasal spray.

Opioids should be used as a last resort (rescue medication) for severe headache when other measures are ineffective.

Chronic migraines

The same medications used to prevent episodic migraine, including monoclonal antibodies that block CGRP, are used to treat chronic migraine. Also, supporting evidence is strong for [onabotulinumtoxinA](#) and [topiramate](#).

Evidence supports use of neurostimulation for acute treatment and prevention of chronic migraine headaches. Noninvasive options include supraorbital stimulation, vagus nerve stimulation, single-pulse transcranial magnetic stimulation, and remote electrical stimulation.

Neuromodulatory treatments

Neuromodulatory treatments can be used to treat and to prevent attacks. These treatments affect brain activity through electrical currents or magnetic fields and can be delivered noninvasively, using commercially available devices.

Noninvasive transcranial magnetic stimulation, using a handheld device applied to the back of the head may relieve acute migraine (4). A device that uses an armband to deliver a nonpainful electrical skin stimulus (called remote electrical neuromodulation) can relieve acute migraine pain. A handheld device that delivers noninvasive [vagus nerve stimulation](#) is also effective.

Trigeminal nerve stimulation, with a device applied to the forehead, can be used in adult patients (≥ 18 years) to treat acute migraine attacks (with or without an aura) or to reduce the frequency of attacks.

Noninvasive neuromodulatory devices have no significant adverse effects. Invasive neuromodulatory treatments are available only at specialized centers and carry greater risks than noninvasive treatments.

Treatment references

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3. [Qaseem A, Tice JA, Etzeandia-Ikobaltzeta I, et al](#). Pharmacologic Treatments of Acute Episodic Migraine Headache in Outpatient Settings: A Clinical Guideline From the American College of Physicians. *Ann Intern Med* 2025;178(4):571–578. doi:10.7326/ANNALS-24-03095
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5. [Lipton RB, Dodick DW, Silberstein SD, et al](#). Single-pulse transcranial magnetic stimulation for acute treatment of migraine with aura: A randomised, double-blind, parallel-group, sham-controlled trial. *Lancet Neurol*. 9:373–380, 2010. doi: 10.1016/S1474-4422(10)70054-5

Prognosis for Migraine

For some patients, migraine is an infrequent, tolerable inconvenience. For others, it is a devastating disorder resulting in frequent periods of incapacity, loss of productivity, and severely impaired quality of life. In a study from a U.S. database (the National Hospital Ambulatory Medical Care Survey and the National Health Interview Survey), migraine was the fifth most common reason for an emergency department visit and 40 percent of adults with migraine were unemployed and economically disadvantaged ([1](#)).

Prognosis reference

1. [Burch R, Rizzoli P, Loder E](#). The prevalence and impact of migraine and severe headache in the United States: Updated age, sex, and socioeconomic-specific estimates from government health surveys. *Headache*. 2021;61(1):60-68. doi:10.1111/head.14024

Prevention of Migraine

Daily preventive therapy is warranted when frequent migraines interfere with activity despite acute treatment.

For patients who use analgesics frequently (eg, > 2 days/week), particularly those with medication overuse headache, preventive drugs should be combined with a program for stopping overused analgesics. Choice of medication can be guided by coexisting disorders, as for the following ([1,2](#)):

- A bedtime dose of [amitriptyline](#) for patients with insomnia
- A beta-blocker for patients with anxiety or coronary artery disease
- Valproate, divalproex, or [venlafaxine](#) for
- An oral [calcitonin](#) gene-related peptide (CGRP)-receptor antagonist ([atogepant](#), [rimegepant](#)) or an injectable CGRP monoclonal antibody (eg, [erenumab](#), [fremanezumab](#), [galcanezumab](#)) if other medications are ineffective
- [Topiramate](#), which can induce weight loss, for patients with obesity or for patients who wish to avoid weight gain

Some experts recommend injection of onabotulinum A as a treatment of choice for chronic migraine prevention using the PREEMPT* Protocol ([3](#)). Neuromodulatory treatments can also help.

Transcutaneous supraorbital nerve stimulation, using a device applied to the forehead, can reduce the frequency of migraines ([4](#)). Transcranial magnetic stimulation, using a device applied to the back of the skull, is indicated for the acute and prophylactic treatment of migraine headache in adolescents (≥ 12) and adults.

Prevention references

1. [Qaseem A, Cooney TG, Etzeandía-Ikobaltzeta I, et al](#). Prevention of Episodic Migraine Headache Using Pharmacologic Treatments in Outpatient Settings: A Clinical Guideline From the American College of Physicians. *Ann Intern Med* 2025;178(3):426-433. doi:10.7326/ANNALS-24-01052

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Key Points

- Migraine is a common primary headache disorder.
- Symptoms can include throbbing unilateral or bilateral pain, nausea, sensitivity to sensory stimuli (eg, light, sounds, smells), nonspecific prodromal symptoms, and temporary neurologic symptoms that precede headache (auras).
- Diagnose migraine based on clinical findings; if patients have red flag findings, imaging and other tests are often needed.
- Involve patients in their care, including avoiding triggers and using biofeedback, stress management, and psychotherapy as appropriate.
- Treat most headaches with analgesics or triptans.
- If attacks are frequent and interfere with activities, use preventive therapy (eg, [amitriptyline](#), a beta-blocker, divalproex, monoclonal antibodies that block [calcitonin](#) gene-related peptide [CGRP], a gepant, or [topiramate](#)), or sometimes injections of [onabotulinumtoxinA](#), or neuromodulatory treatments.

Drug Information for the Topic



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